

Article

Design and Experimental Validation of a Programmable Pressure Pneumatic Knife CNC Platform for Flexible Sheet Cutting

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Abstract

This study presents a programmable pressure pneumatic knife CNC platform for the fabrication and experimental investigation of flexible sheet materials. The platform enables independent control of blade-loading pressure, feed rate, and toolpath motion, allowing pressure–feed rate operating windows for mechanical knife cutting to be established. Paper and vinyl sheets are evaluated using square and straight-line cutting patterns, with cutting quality classified into four regimes: clean cut, skimming, tearing, and crushing. The results demonstrate that insufficient pressure causes unstable blade–material contact, resulting in skimming or no-touch behavior, whereas excessive pressure promotes tearing. Paper attains a 99% clean cut rate at 1 bar and 9000 mm min^{−1} during square cutting, while vinyl exhibits 100% clean cuts across 3000 mm min^{−1} to 9000 mm min^{−1} at 3 bar. Straight-line cutting achieves clean cut rates of 98% for paper and 97% for vinyl at 2 bar and 7000 mm min^{−1}. These findings support the effectiveness of programmable pressure mechanical knife cutting for repeatable fabrication of flexible sheet materials.

Keywords: knife CNC machine; flexible sheet cutting; cyber-physical machines; computer-aided manufacturing (CAM); computer-integrated manufacturing (CIM)

1. Introduction

Flexible sheet materials are increasingly employed in soft robotics, wearable assistive systems, rehabilitation devices, flexible sensors, medical patches, pneumatic structures, and compliant human–machine interfaces [1–3]. These applications commonly rely on polymer films, adhesive tapes, textile-backed sheets, elastomers, foams, and paper-based substrates that must be converted into precise geometries before bonding, sealing, assembly, or actuation [4]. Cutting quality directly influences dimensional accuracy, mechanical response, sealing integrity, repeatability, and device reliability.

Among the principal methods used to process flexible textiles and sheet materials are laser, ultrasonic knife, and mechanical knife cutting [5–7]. Laser cutting provides non-contact processing, high precision, and excellent suitability for intricate geometries, but can cause melting, charring, discoloration, and process emissions [5]. Ultrasonic knife cutting reduces cutting resistance through high-frequency blade vibration and is effective for technical and reinforced textiles. However, it requires specialized hardware and careful resonance tuning. Mechanical knife cutting is a cold process with broad



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material compatibility, low equipment complexity, and straightforward integration into CNC platforms. Despite these advantages, its performance remains sensitive to blade loading, feed rate, tool condition, and workpiece fixation [8–10].

Flexible sheet cutting differs fundamentally from conventional machining because the workpiece can stretch, wrinkle, compress, lift, adhere to the cutting surface, or tear before complete separation [9,11]. The cutting response is therefore governed not only by the programmed toolpath but also by the coupled effects of normal blade loading, feed motion, blade orientation, material compliance, interfacial adhesion, fixture condition, and cutting board flatness. Insufficient loading can produce skimming or incomplete penetration, whereas excessive loading can increase frictional drag, wedging resistance, material deformation, and uncontrolled tearing. These competing effects suggest that satisfactory cutting can only be achieved within a bounded operating region [8,10,12]. The boundaries of this operating region are material- and geometry-dependent. Contact-based cutting remains widely used for flexible sheet materials because of its simple configuration and direct mechanical interaction. Manual knife cutting, however, depends strongly on operator skill and applied force, while conventional drag knife plotters provide automated tool motion with limited blade-loading control. Active force-controlled CNC systems can improve control of the cutting force but require additional force sensing and feedback hardware. Non-contact alternatives such as laser cutting offer high positional accuracy but may cause thermal damage to heat-sensitive materials, whereas ultrasonic cutting reduces cutting resistance but requires specialized equipment and tooling. The main challenges and distinctive features of these cutting methods are summarized in Table 1.

Table 1. Comparison of cutting methods for flexible sheet materials.

Cutting Method	Main Challenges for Flexible Sheets	Distinctive Features	References
Manual knife cutting	Cutting performance depends on operator skill and applied force, resulting in limited repeatability	Simple operation and low equipment cost	[8–12]
Conventional drag knife plotter	Limited control of blade loading; cutting performance is sensitive to material properties and toolpath conditions	Automated toolpath control with a simple mechanical cutting mechanism	[6,7]
Active force-controlled CNC cutting	Requires force sensing and feedback control, increasing system complexity	Direct regulation of blade–material interaction force	
Laser cutting	Potential thermal damage, melting, or discoloration of heat-sensitive materials	Non-contact processing with high positional accuracy	[5]
Ultrasonic cutting	Requires specialized equipment and tooling; process conditions must be matched to material characteristics	High-frequency blade vibration reduces cutting resistance	[6]
Proposed pneumatic knife CNC	Blade-loading force depends on pneumatic and mechanical conditions and requires pressure-to-force calibration	Programmable blade loading integrated with CNC-controlled tool motion	Present study; [8–10]

The novelty of this study is the integration of programmable pneumatic blade loading with CNC-controlled tool motion, allowing blade loading and feed rate to be systematically

varied as independent process parameters. This approach is used to characterize material- and geometry-dependent operating windows for reliable flexible sheet cutting.

Fragile paper-based sheets are susceptible to tearing and local buckling. Adhesive polymer tapes must overcome elastic recovery and interfacial resistance. Compressible foams deform before separation, and cloth-backed tapes exhibit increased cutting resistance due to fiber reinforcement [9,13]. Toolpath features such as corners, curves, internal holes, and closely spaced cuts further increase the requirement for blade alignment and material stability. A cutting condition that is effective for one material or geometry cannot be assumed to remain suitable for another [10].

For low-volume fabrication, manual cutting and commercial craft cutters provide practical solutions. Manual operation is affected by continuous variation in hand pressure, blade angle, cutting speed, and path-following accuracy. Commercial drag knife systems improve planar positioning. Nevertheless, they commonly rely on fixed blade depth, empirical presets, or manually adjusted loading. Such configurations provide limited access to the internal control strategy and do not readily support independent variation in feed rate, tool position, and blade-loading pressure. As a result, quantitative investigation of blade–material interaction and material-specific cutting windows remains difficult [4,7]. However, a systematic experimental framework that independently varies blade-loading pressure and feed rate while linking these parameters to distinct cutting quality regimes across flexible sheet materials remains insufficiently established. The novelty of this study lies in treating programmable pneumatic blade loading as an independent CNC process variable and using it, together with feed rate, to establish material- and geometry-dependent operating windows for reliable flexible sheet cutting.

To address these gaps, this study develops and experimentally validates a programmable pressure pneumatic knife CNC platform for process window-controlled flexible sheet cutting. The system combines a Cartesian motion stage, a self-aligning pneumatic knife toolhead, an electro-pneumatic regulator, solenoid-based knife engagement, and Klipper-based distributed control [14]. Planar motion, Z-axis positioning, pneumatic pressure, and knife engagement are prescribed independently within a unified fabrication workflow. The commanded pressure is additionally converted into nominal blade-loading force to support mechanical interpretation of the cutting process. This integrated architecture enables programmable pneumatic blade loading with CNC motion control and systematically establishes pressure–feed rate operating windows that relate blade loading conditions to cutting quality regimes across different flexible sheet materials and toolpath geometries. The platform is further evaluated through material-specific cutting trials, manual–machine comparisons, and fabrication demonstrations involving a kirigami gripper, flexible pneumatic bag, and silicone scar sheet.

2. System Design and Operating Principles

2.1. Overall System Architecture

The platform was developed as a programmable pressure pneumatic knife CNC system for reproducible and controlled cutting of flexible sheet materials. As shown in Figure 1, the system comprises five main subsystems: a Cartesian CNC motion stage, a pneumatic knife toolhead, a pressure regulation and actuation unit, a Klipper-based distributed control system, and a user-facing web interface. The architecture decouples planar toolpath motion from vertical blade-loading control, allowing the cutting trajectory, feed rate, Z-axis position, pneumatic pressure, and knife engagement to be controlled independently within a unified CNC workflow. The mechanical, pneumatic, and control architectures of these subsystems are described in the following sections [8,9].

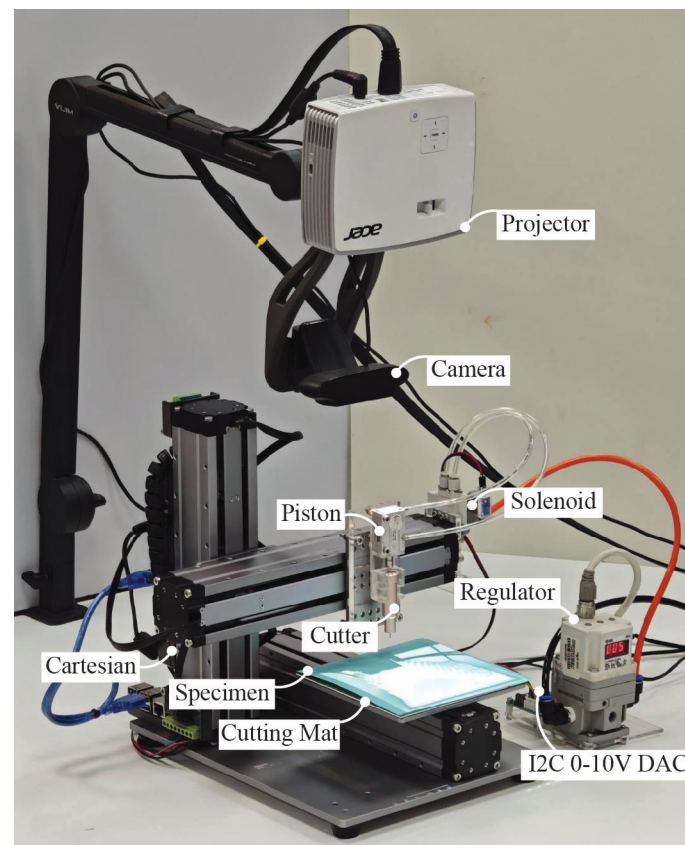


Figure 1. Overall system architecture.

The Cartesian CNC stage provides the primary motion capability of the system. The X–Y axes define the programmed cutting trajectory and feed rate, while the Z-axis sets the initial tool position relative to the material surface. In addition to this positional control, pneumatic blade loading provides a controllable interaction between the blade and workpiece. This configuration is particularly important for flexible sheets, where variations in material thickness, surface flatness, fixture condition, and cutting board wear can significantly affect the blade–material contact condition [9,10].

The pneumatic subsystem consists of an electro-pneumatic pressure regulator, a solenoid valve, and a pneumatic actuator integrated with the knife toolhead. The electro-pneumatic regulator determines the commanded blade-loading pressure, whereas the solenoid valve controls knife engagement and disengagement during the cutting sequence. In this arrangement, pneumatic pressure is treated as a programmable process variable rather than a manually adjusted setting. This architecture enables the system to investigate how blade-loading pressure interacts with feed rate, material compliance, adhesion, and toolpath geometry to produce different cutting outcomes [8,12].

The control architecture is implemented using a Klipper-based distributed control framework. In this configuration, high-level G-code interpretation, macro execution, and user interface communication are handled by the host computer. Time-critical motion and input/output (I/O) operations are executed by the microcontroller [14]. This separation facilitates coordinated control of CNC motion, Z-axis positioning, pneumatic pressure control, and solenoid actuation. As a result, the cutting process can be programmed, repeated, and modified systematically without relying on manual intervention during operation.

The user interface provides access to the machine control workflow, including toolpath execution, pressure setting, feed rate selection, and actuation commands. This platform functions not only as a fabrication device, but also as an experimental system for process

window characterization. By integrating open CNC motion control with programmable pneumatic blade loading, the architecture provides a flexible machine design framework for studying and optimizing knife cutting of compliant sheet materials. This integrated system forms the basis for the subsequent pressure–feed rate operating window experiments and material-specific cutting trials.

2.2. Pneumatic Knife Mechanism

The pneumatic knife mechanism was designed to provide controlled blade engagement and programmable blade loading in a compact and modular structure suitable for installation on a Cartesian CNC platform. As shown in Figure 2, the toolhead consists of a replaceable cutting blade, a blade holder, a self-aligning rotational support, a pneumatic actuator, and a mounting interface connected to the Z-axis carriage. The mechanism converts regulated pneumatic pressure into a controllable normal blade-loading condition at the blade–material interface. The CNC motion stage defines the cutting trajectory and feed rate.

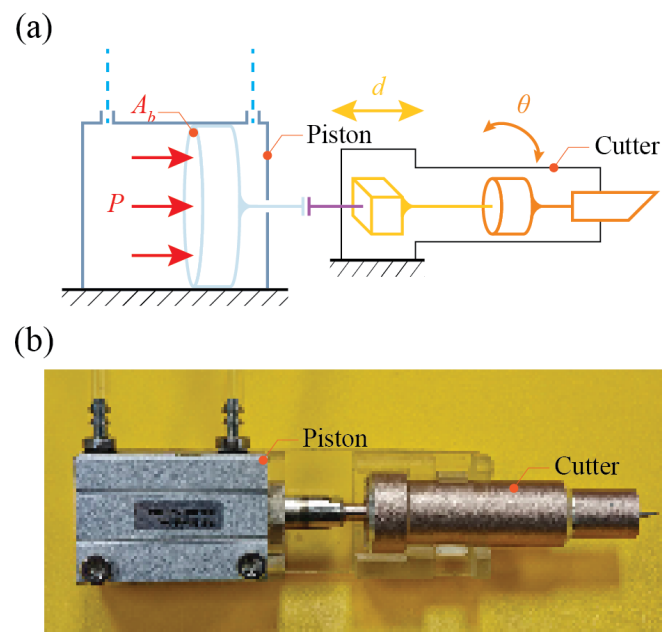


Figure 2. Mechanical design of the programmable pressure pneumatic knife toolhead: (a) Schematic of the motorized pneumatic knife toolhead and its main components, and (b) Photograph of the fabricated toolhead.

A key feature of the toolhead is the self-aligning blade mechanism, which allows the blade to passively rotate about its support axis so that the cutting edge can align with the instantaneous feed direction. This alignment reduces lateral resistance and helps the blade follow curved and multi-directional toolpaths while minimizing material dragging, stretching, and tearing [9,11]. The blade is also replaceable, allowing different blade geometries or worn cutting edges to be exchanged according to the material being processed. This modularity supports consistent cutting conditions across materials with different thickness, stiffness, adhesion, compressibility, and reinforcement characteristics and reduces the influence of blade wear on experimental results [9,13].

Vertical blade loading is generated by the pneumatic actuator. When engaged, regulated air pressure drives the blade toward the material surface and produces a nominal normal cutting force. This loading condition promotes blade–material contact and penetration while accommodating small variations in material thickness, surface flatness, and fixture condition [8,10]. Unlike a fixed-depth drag knife, the pneumatic mechanism pro-

vides a compliant loading mode that maintains blade contact without requiring a fixed cutting depth [7]. The pressure magnitude and knife engagement are controlled independently: the pneumatic pressure determines the nominal blade-loading force, while the solenoid valve controls knife engagement and disengagement during the cutting sequence. This separation allows pressure to be predefined and knife activation to be synchronized with the programmed toolpath.

Overall, the pneumatic knife mechanism combines controlled normal loading, passive directional alignment, and replaceable blade mounting to support repeatable mechanical cutting of flexible sheet materials. These features address the coupled effects of blade penetration, frictional drag, material deformation, and feed motion that influence cutting quality [8–10]. The resulting toolhead provides a practical mechanism for implementing programmable pressure knife cutting within the CNC platform.

2.3. Pressure Regulation and Knife Engagement

The pressure regulation and knife engagement subsystem provides programmable control of blade loading and knife activation during CNC cutting. It consists of an electro-pneumatic pressure regulator (SMC ITV2030, SMC Corporation, Tokyo, Japan), solenoid valve (SMC SY3020, (SMC Corporation, Tokyo, Japan)), pneumatic tubing, a compressed air supply, and a pneumatic actuator (SMC CDUJB10-15D, (SMC Corporation, Tokyo, Japan)) integrated with the knife toolhead. The regulator controls the pressure supplied to the actuator, while a solenoid valve controls knife engagement and disengagement [15].

The electro-pneumatic regulator receives an electrical command from the control system and continuously regulates the supply pressure. This capability allows the blade-loading pressure to be specified digitally and varied systematically during process window experiments. Knife engagement is controlled by a solenoid valve installed between the pressure regulation unit and the pneumatic actuator. When activated, the valve supplies pressurized air to the actuator and engages the knife. When deactivated, the knife is disengaged for travel moves, repositioning, or toolpath transitions without cutting. This configuration synchronizes knife engagement with CNC motion and prevents unwanted blade contact that could scratch, drag, or deform flexible sheets.

The independent control of pressure and engagement improves experimental repeatability by allowing the same pressure condition and activation sequence to be reproduced across different materials, feed rates, and toolpath geometries. This capability is essential for identifying pressure–feed rate operating windows and comparing cutting quality regimes. Overall, the subsystem links programmable pneumatic loading and synchronized knife actuation to the Klipper-based control architecture, as described in the following section.

2.4. Klipper-Based Distributed Control

The platform employs a Klipper-based distributed control architecture that separates high-level process coordination from time-critical machine execution [14]. In Figure 3, the host computer manages G-code interpretation, trajectory coordination, macro execution, parameter configuration, and web interface communication. The microcontroller performs stepper control, end-stop monitoring, digital switching, and synchronized I/O operations. This division enables flexible process programming while maintaining coordinated motion and actuator timing.

Cartesian motion, Z-axis positioning, pneumatic pressure commands, and solenoid-based knife engagement are coordinated through configurable G-code macros. Each macro defines the cutting sequence, including tool positioning, pressure assignment, knife engagement, toolpath execution, disengagement, and retraction. Consequently, pressure and feed

rate can be varied systematically without modifying the firmware or machine hardware, thereby improving experimental repeatability during process window characterization.

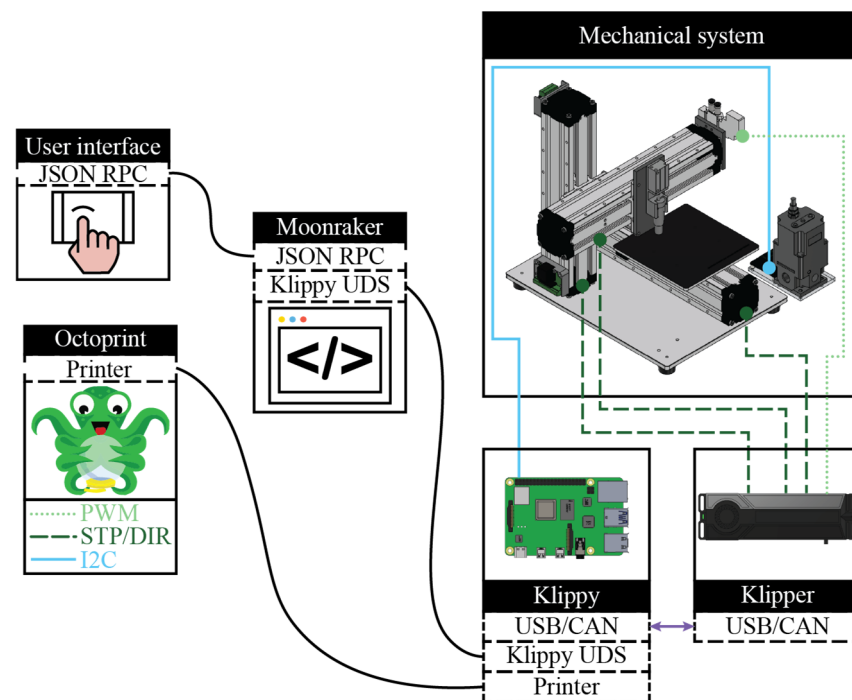


Figure 3. Klipper-based control system architecture and web interface. The diagram illustrates the communication links between the host (Klipper / Moonraker), mainboard controller (Klipper MCU), mechanical CNC hardware, and web interfaces (User Interface/OctoPrint).

The architecture also provides communication links among the host controller, microcontroller, machine sensors, pressure regulation unit, CNC motion system, and web interface. The web interface enables toolpath execution, parameter adjustment, machine status monitoring, and modification of cutting conditions. The open Klipper framework provides a reproducible and extensible control environment for programmable pressure knife cutting and future integration of process-monitoring and feedback control functions.

2.5. Hybrid Position–Force Cutting Concept

The platform adopts a hybrid position–force cutting concept in which planar motion and blade loading are controlled through separate but coordinated mechanisms, as illustrated in Figure 4 [16,17]. The Cartesian CNC stage controls the X–Y cutting trajectory, feed rate, and Z-axis tool position, while the pneumatic knife mechanism provides programmable normal blade loading. This arrangement decouples toolpath position from blade–material interaction, allowing both parameters to be adjusted independently within the same fabrication workflow.

The need for this separation arises from the sensitivity of flexible sheet cutting to blade loading. Cutting performance is strongly affected by the applied blade loading. Insufficient loading may cause skimming or incomplete penetration, whereas excessive loading can increase frictional drag, material deformation, and tearing. The pneumatic actuator also provides compliant loading that can accommodate small variations in material thickness, surface flatness, fixture condition, and cutting board height. Thus, feed rate and pneumatic pressure can be treated as independent process variables for different flexible sheet materials and toolpath geometries.

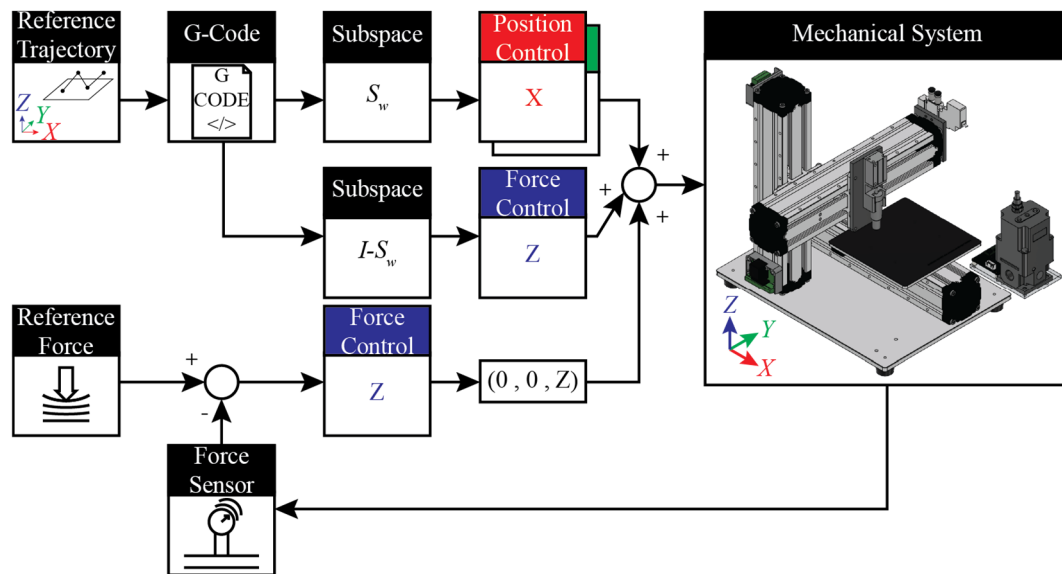


Figure 4. Overall architecture of the hybrid position–force-controlled programmable pressure pneumatic knife CNC platform.

This architecture supports pressure–feed rate operating window characterization, with cutting outcomes classified into clean cut, tearing, skimming, and crushing regimes. The resulting framework allows the cutting response to be evaluated systematically rather than through empirical trial and error. The term “force” refers to the nominal blade-loading force estimated from the commanded pneumatic pressure. The platform does not employ closed-loop force feedback during cutting. Instead, the system provides pressure-programmed blade loading combined with CNC position control rather than direct force-controlled cutting.

3. Pressure to Force Relationship and Cutting Mechanics

3.1. Cutting Quality Regimes and Operating Window

Mechanical knife cutting of flexible sheets is governed by the coupled effects of blade loading, material resistance, friction, wedging, feed motion, and local deformation. In Figure 5, the pneumatic actuator generates a nominal normal force that promotes blade penetration, while the advancing blade develops shear, frictional, wedging, and dynamic forces at the blade–material interface [8,11,18]. Because flexible sheets can deform or move during cutting, the cutting outcome also depends on the blade–material interaction and the programmed toolpath [9,10,12]. In the platform, pneumatic pressure provides programmable blade loading, while the CNC system independently controls the feed direction and feed rate.

The cutting model is formulated using parameters describing the material properties, blade geometry, and loading characteristics, as summarized in Table 2. The variables employed in the model and their definitions are listed in Table 3.

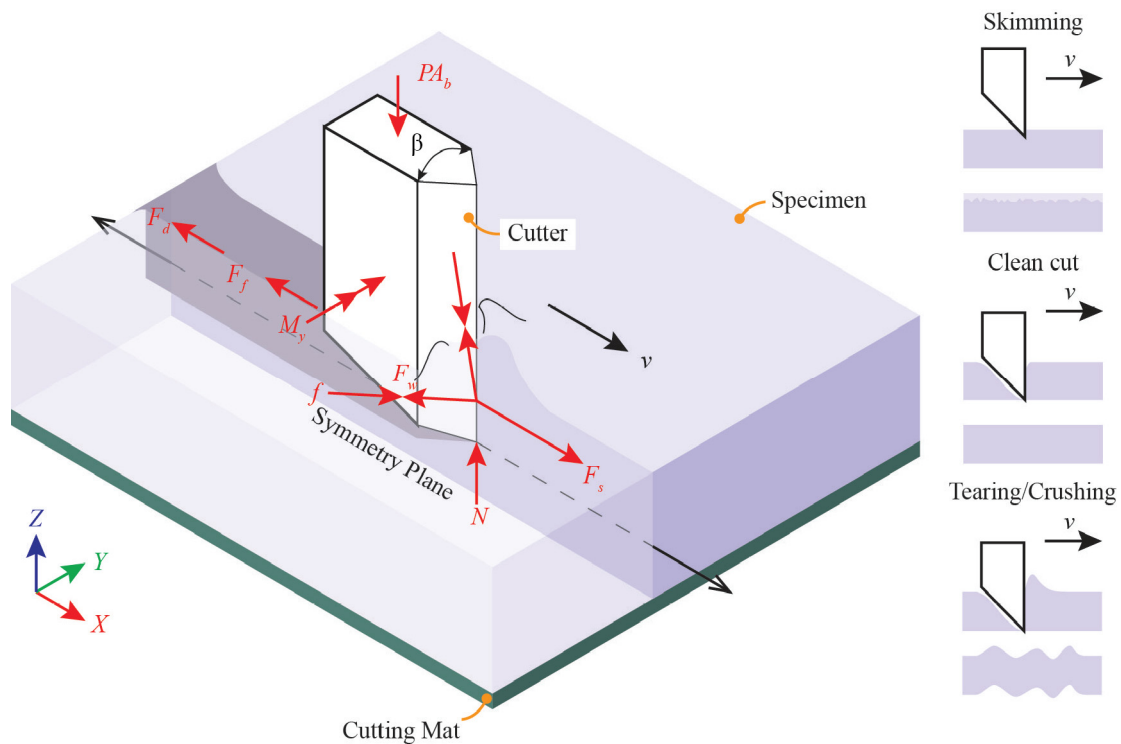


Figure 5. Cutting principle of the programmable pressure pneumatic knife. The blade applies a controlled normal force while advancing in the programmed feed direction. The figure illustrates the principal cutting forces and the resulting cutting conditions, including skimming, clean cutting, and tearing/crushing.

Table 2. Parameters.

Symbol	Definition	Unit
τ_c	Material shear strength	Pa
σ_c	Material compressive strength	Pa
$F_{\text{tear},c}$	Critical tearing resistance	N
μ	Blade–material coefficient of friction	–
A	Effective piston area	m ²
A_s	Effective shear area	m ²
A_c	Blade–material contact area	m ²
α	Angular acceleration	rad/s ²
β	Blade included (wedge) angle	°
K_v	Dynamic cutting coefficient	N/(m/s) ^m
m	Feed rate exponent	–
K_b	Compression–buckling coefficient	m/s
ϵ	Regularization velocity constant	m/s

Table 3. Variables.

Symbol	Definition	Unit
P	Applied pneumatic pressure	bar
F	Pneumatic cutting force	N
N	Normal (downward) cutting force	N
F_s	Effective shear force at the blade edge	N
F_f	Frictional drag force between the blade and the material	N
F_w	Wedging force generated by the blade geometry	N
F_d	Dynamic cutting force due to feed rate effects	N

Table 3. Cont.

Symbol	Definition	Unit
f	Friction force	N
v	Feed rate (cutting speed)	mm min ⁻¹
S	Shear Index	–
D	Drag–Tear Index	–
CB	Compression–Buckling Index	–

The following equations describe the cutting mechanics.

Pressure-to-Force. The relationship between applied pneumatic pressure and generated cutting force can be expressed as [19,20]:

$$F = PA \quad (1)$$

Moment about the Y-axis. The rotational dynamics about the Y-axis can be described by [21]:

$$\Sigma M_y = I\alpha_y \quad (2)$$

Shear Index. The shear index represents the ratio between the effective cutting shear force and the material shear resistance:

$$S = \frac{F_s}{A_s T_c} \quad (3)$$

Drag–tear Index. The drag–tear index evaluates the influence of friction, wedging, and dynamic cutting forces relative to the critical tearing resistance:

$$D = \frac{F_f + 2F_w \cos\left(\frac{\beta}{2}\right) - 2f \sin\left(\frac{\beta}{2}\right) + F_d}{F_{tear,c}} \quad (4)$$

where the dynamic cutting force is defined as [22,23]:

$$F_d = K_v v^m \quad (5)$$

Compression–Bunching Index. The compression–bunching index represents the tendency for material accumulation caused by insufficient feed motion:

$$CB = \frac{K_b N}{A_c \sigma_c (v + \epsilon)} \quad (6)$$

3.2. Pressure-to-Force Conversion

The commanded pneumatic pressure was converted to nominal normal blade-loading force using the pressure–area relationship defined in Section 3.1 [20,24].

Since the pressure command was specified in bar, the conversion can be written as:

$$F_N = (P \times 10^5)A \quad (7)$$

where F_N is the nominal normal blade-loading force in newtons, P is the commanded pressure in bar, and A is the effective piston area in m² [25,26].

For the pneumatic actuator used in this study:

$$A = 78.5 \times 10^{-6} \text{ m}^2 \quad (8)$$

Substituting this value into Equation (8) gives:

$$F_N = 7.85P \quad (9)$$

Thus, each 1 bar increase in pressure corresponds to approximately 7.85 N of nominal normal blade-loading force.

Equation (9) was used throughout the experiments to convert the commanded pneumatic pressure into the corresponding nominal blade-loading force. As summarized in Table 4, the nominal force calculated from the pressure–area relationship was compared with the experimentally measured force obtained from five repeated measurements at each pressure level. As the commanded pressure increased from 0.5 to 3.5 bar, the measured force increased from 3.39 to 25.75 N, while Equation (9) predicted nominal values from 3.93 to 27.48 N. The measured forces were consistently slightly lower than the nominal values, with deviations ranging from 0.21 to 1.73 N, indicating modest pneumatic and mechanical losses not represented by the ideal pressure–area model. The corresponding standard deviations ranged from 0.12 to 2.31 N. These results demonstrate that Equation (9) provides an appropriate first-order conversion for defining the blade-loading conditions used in the cutting experiments, while the measurements reported in Table 4 quantify the difference between nominal and actual applied force.

Table 4. Conversion between pneumatic pressure and nominal blade-loading force.

Pressure (Bar)	Measured Force (N)	Nominal Blade-Loading Force (N)	Standard Deviation (N)
0.5	3.39	3.93	0.28
1.0	7.64	7.85	0.47
1.5	11.20	11.78	0.16
2.0	14.47	15.70	1.13
2.5	18.89	19.63	0.12
3.0	22.76	23.55	0.44
3.5	25.75	27.48	2.31

3.3. Cutting Quality Regimes

Cutting quality was classified into four regimes: clean cut, tearing, skimming, and crushing. This classification was used to characterize the dominant blade–material interaction and to evaluate the effects of pneumatic pressure, nominal blade-loading force, feed rate, material properties, and toolpath geometry on cutting performance.

A clean cut was defined as complete penetration and separation of the material along the programmed toolpath with minimal edge damage and no residual connection. This condition represents the desired cutting regime and indicates sufficient blade loading with an appropriate feed rate [8–10].

Tearing refers to uncontrolled material rupture or edge damage occurring along or adjacent to the intended cutting path. It is associated with excessive blade loading, friction, wedging effects, or unstable material deformation during cutting [11,12].

Skimming describes incomplete cutting in which the blade scratches or scores the material surface without full penetration. This condition generally occurs when blade loading is insufficient or when the effective cutting depth is reduced by material or surface variations [7,9].

Crushing describes excessive compressive interaction between the blade and material, resulting in substantial local deformation rather than stable cutting. This regime is associated with excessive blade loading and material compression [7,26].

At low blade loading, cutting tends to shift toward skimming, whereas excessive loading increases the likelihood of tearing or crushing. Stable clean cutting is obtained within an appropriate pressure–feed rate window that depends on the material and toolpath geometry.

For quantitative classification, the cutting outcome was determined from the following criteria:

Skimming:	$S < 1$
Clean Cut:	$S \geq 1, D < 1, CB < 1$
Tearing:	$S \geq 1, D \geq 1, CB < 1$
Crushing:	$S \geq 1, D < 1, CB \geq 1$

4. Materials and Methods

This section describes the experimental procedure used to evaluate the programmable pressure pneumatic knife CNC platform for flexible sheet cutting. Representative materials were selected to cover different cutting characteristics, including paper sheets, polymer films, adhesive tapes, foam, textile-reinforced tape, and silicone elastomer sheets. The platform controlled the trajectory, feed rate, Z-axis position, pneumatic pressure, and knife engagement within a unified Klipper-based CNC workflow. Pressure–feed rate experiments were first conducted on paper and vinyl sheets to characterize cutting quality transitions, followed by material-specific trials to assess adaptability to different substrate properties. A manual–machine comparison was performed to evaluate processing time, trajectory consistency, and visible defect reduction. Finally, demonstration cases involving a kirigami gripper, a flexible pneumatic bag, and a silicone scar sheet were fabricated to demonstrate the applicability of the platform to soft robotic and wearable device applications [1–3,27].

4.1. Experimental Setup

In Figure 6, the experimental setup comprised the Cartesian CNC stage, programmable pressure pneumatic knife, electro-pneumatic regulator, solenoid valve, Klipper controller, cutting board, and specimen fixture. The X–Y axes executed the prescribed toolpath, while the Z-axis established the initial blade position. The regulator controlled pneumatic pressure for blade loading, and the solenoid valve controlled knife engagement. A consistent fixation procedure was used to minimize the sliding, lifting, and wrinkling of the material. The commanded pneumatic pressure was converted to nominal blade-loading force using the relationship defined in Section 3.

After cutting, the samples were photographed and classified into clean cut, skimming, tearing, and crushing according to Section 3.3. Processing time and trajectory quality were recorded for the manual–machine comparison.

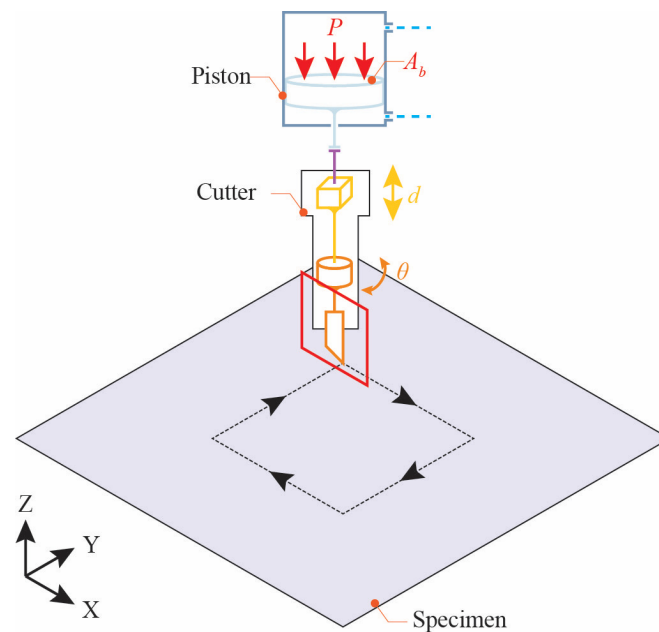


Figure 6. Experimental setup, pressure–feed rate operating windows and cutting performance.

4.2. Pressure–Feed Rate Experiments

Pressure–feed rate experiments were conducted on paper and vinyl using straight-line and square toolpaths [8]. The straight-line geometry isolated blade–material interaction under a constant cutting direction. The square geometry introduced repeated direction changes and passive blade reorientation [4,28]. For square cutting, pneumatic pressure varied from 0 to 3 bar and the feed rate from 3000 to 9000 mm min^{−1}, corresponding to nominal blade-loading forces of 0–23.55 N. For straight-line cutting, two parameter sweeps were performed: pressure varied from 0 to 4 bar at a fixed feed rate of 6000 mm min^{−1}, and feed rate varied from 1000 to 9000 mm min^{−1} at a fixed pressure of 2 bar. Each straight cut was 100 mm long, with adjacent paths spaced 20 mm apart. The clean cut ratio was used as the primary response variable, while the remaining categories were recorded to characterize the corresponding failure modes. The resulting data were used to construct material- and geometry-specific pressure–feed rate operating maps [4].

4.3. Material-Specific Experiments

Material-specific trials were conducted to assess the applicability of the platform to five flexible sheet substrates with distinct structures: PVC electrical tape, LDPE foam, masking tape, polyimide film tape, and cloth-backed tape. These materials represented elastic adhesive film, compressible foam, paper-based adhesive sheet, thin engineering film, and fiber-reinforced tape, respectively. The thicknesses and representative mechanical properties of the materials used in the experiments are summarized in Table 5.

A representative pressure–feed rate condition was identified for each material through preliminary tuning rather than full operating window mapping. PVC electrical tape was cut at 3.5 bar and 3500 mm min^{−1}; LDPE foam and masking tape at 2 bar and 5000 mm min^{−1}; polyimide tape at 1.5 bar and 6000 mm min^{−1}; and cloth-backed tape at 3 bar and 3000 mm min^{−1}. The corresponding nominal blade-loading forces were calculated using the experimental pressure-to-force relationship defined in Section 3.2.

Table 5. Material properties and specimen thicknesses used in the cutting experiments.

Material	Thickness (mm)	Tensile Strength (MPa)	Young's Modulus (GPa)	Tear Strength
Paper	0.10	30–50	2–4	Low
Vinyl	0.08	15–30	1.5–3.0	Moderate
PVC electrical tape	0.12	15–20	0.1–0.5	Moderate to high
LDPE foam sheet	0.03	0.2–0.5	0.01–0.05	Low to moderate
Masking tape	0.12	30–45	1.5–2.5	Low
Polyimide tape	0.05	140–230	2.5–3.5	High
Cloth tape	0.025	50–80	0.5–1.0	Direction-dependent
Silicone sheet	0.57	5–10	0.001–0.05	Moderate

Note: Specimen thicknesses were measured experimentally. The mechanical property values are representative ranges obtained from the relevant material specifications and literature. Tear strength and compression behavior are reported qualitatively because quantitative measurements were not performed in this study.

4.4. Manual–Machine Cutting Comparison

Manual and machine cutting were compared using the same material and nominal S-shaped trajectory, which introduced continuous curvature and changing cutting direction. The manual sample was cut by an operator using a hand-held knife. In comparison, the machine sample was produced using the programmable pressure CNC platform at prescribed pressure and feed rate settings [4,29]. Processing time was recorded for both methods. The resulting cut traces were photographed and evaluated for trajectory continuity, groove width uniformity, surface damage, tearing, and local deviation [4]. An image-based overlay was also generated by extracting and aligning the two trajectories to support qualitative comparison of path consistency [30]. Cutting defects were classified according to the regimes defined in Section 3.3 [7].

4.5. Application Demonstration

Three application-oriented geometries were fabricated: a kirigami gripper, a flexible pneumatic bag, and a silicone breast scar sheet [2,3,27]. The kirigami pattern contained repeated narrow slits and was cut at 3 bar and 3000 mm min^{−1}. The pneumatic bag pattern included closed contours, internal features, and small-radius curves and was cut at 3 bar and 3000 mm min^{−1}. The silicone scar sheet profile contained continuous curved paths and was cut at 3 bar and 2000 mm min^{−1}. All toolpaths were generated digitally and executed using the Klipper-based workflow described in Section 2.4 [14]. Pneumatic pressure was set before each operation, and knife engagement was synchronized with the cutting path through the solenoid valve. The fabricated samples were visually inspected for cut continuity, complete separation, edge damage, residual connections, and geometric distortion.

5. Results

5.1. Straight-Line Cutting on Paper and Vinyl

In Figure 7, the straight-line cutting results for paper and vinyl are presented. Both materials exhibited distinct pressure–feed rate conditions for stable cutting.

For paper, moderate pressure improved blade penetration and cutting stability, whereas excessive pressure promoted tearing. At 2 bar, increasing the feed rate improved cutting performance up to a maximum clean cut ratio of 98% at 7000 mm/min. Increasing the feed rate further to 9000 mm/min caused a sharp reduction in cutting quality, indicating insufficient penetration stability at high speed.

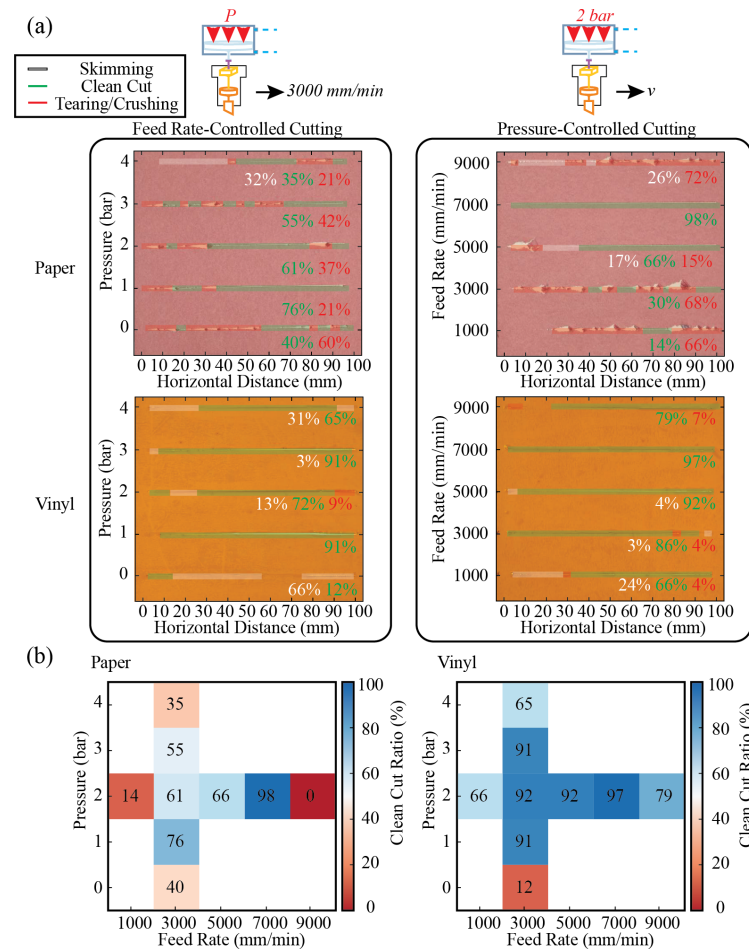


Figure 7. Straight-line cutting results for paper and vinyl under different pressure and feed rate conditions: (a) Experimental cutting results for feed rate-controlled and pressure-controlled cutting, showing the resulting cutting conditions, and (b) Clean cut ratio (%) summarized as a function of pressure and feed rate.

For vinyl, sufficient pneumatic loading was required to achieve stable blade contact. Without pneumatic loading, skimming and incomplete cuts dominated, while increasing pressure improved cutting quality. At 2 bar, the highest clean cut ratio of 97% was obtained at 7000 mm/min, with a reduction at 9000 mm/min. However, vinyl remained more tolerant of high feed rates than paper.

Overall, both materials achieved their highest clean cut ratios at 2 bar and 7000 mm/min, with 98% for paper and 97% for vinyl. Paper was more susceptible to tearing and exhibited a narrower operating window. Vinyl required adequate blade loading and maintained stable cutting over a wider range of feed rates. These results confirm that stable straight-line cutting depends on balancing blade loading and feed rate.

The straight-line cutting experiments on paper and vinyl were repeated to assess the repeatability of the identified cutting conditions. For each condition, three independently prepared samples were tested. The experimental setup was also refined to improve specimen fixation and alignment, and a new knife blade was installed to minimize variability associated with blade wear. Under these improved and controlled conditions, clean and complete cuts were consistently obtained at the identified operating conditions for both materials. The repeated trials therefore confirm that the previously identified parameter settings provide reproducible cutting performance when the machine setup, blade condition, and specimen preparation are appropriately controlled.

5.2. Square Cutting on Paper and Vinyl

In Figure 8, the square cutting results for paper and vinyl are presented. Compared with straight-line cutting, square cutting was more sensitive to pneumatic pressure and feed rate because the blade repeatedly changed direction at the corners.

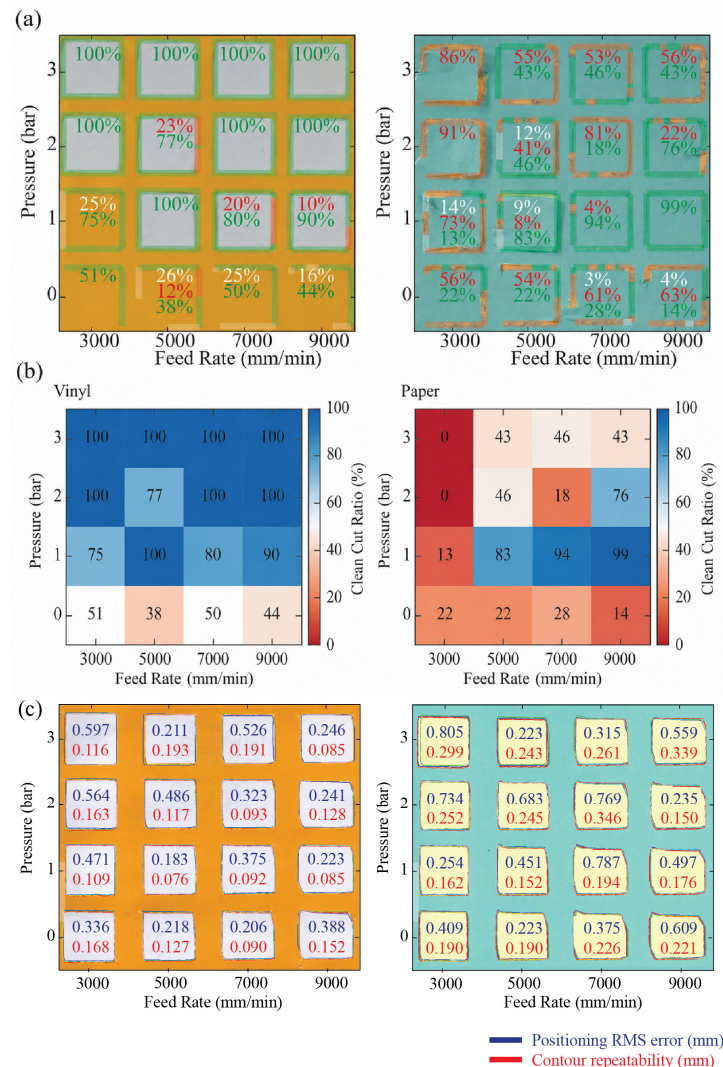


Figure 8. Square-cutting performance of paper and vinyl under different pressure and feed rate conditions: (a) Experimental cutting results under varying pressure and feed rate, (b) Heat maps of the corresponding clean cut ratio (%) as a function of pressure and feed rate, and (c) Positioning and repeatability performance evaluated from repeated square-cutting experiments, reported in terms of positioning RMS error (mm) and contour repeatability (mm). Green indicates clean cutting, yellow indicates skimming, red indicates tearing, and white indicates no contact.

For paper, insufficient pressure resulted in no-touch and incomplete cuts. The best performance was obtained at 1 bar, with a maximum clean cut ratio of 99% at 9000 mm/min. Increasing the pressure beyond 1 bar reduced cutting quality, with tearing becoming more prevalent, particularly at lower feed rates.

For vinyl, higher pressure was required for stable cutting. The most robust performance was obtained at 3 bar, where all tested feed rates achieved 100% clean cuts.

Overall, paper achieved its best performance at 1 bar and 9000 mm min⁻¹, while vinyl required 3 bar and remained relatively insensitive to feed rate once sufficient pressure was applied. These results show that the optimum square-cutting condition depends strongly

on the material and requires appropriate blade loading to maintain stable penetration without excessive tearing.

The square-cutting experiments were repeated on paper and vinyl using three independently prepared samples for each pressure–feed rate condition. All tested conditions produced complete clean cuts; therefore, these experiments were used specifically to evaluate the positioning accuracy, contour repeatability, and geometric fidelity of the cutting system rather than to distinguish cutting quality regimes. A nominal square was commanded, and the resulting contours were evaluated in terms of fitted side length, positioning error, contour repeatability, perfect square error, and squareness error.

For paper, the fitted side length ranged from 19.462 to 20.378 mm, with positioning errors of 0.223–0.805 mm and contour repeatability values of 0.150–0.346 mm. For vinyl, the fitted side length ranged from 19.694 to 20.221 mm, while the positioning RMS error ranged from 0.183 to 0.597 mm and contour repeatability from 0.076 to 0.193 mm. These results indicate that the commanded dimensions were reproduced consistently for both materials, with vinyl exhibiting lower contour variability under the tested conditions.

Feed rate showed the clearest systematic influence on geometric fidelity for both materials. For paper, the mean perfect square error increased from approximately 0.292 mm at 3000 mm/min to 1.033 mm at 9000 mm/min, while the corresponding mean squareness error increased from approximately 1.11% to 18.79%. Vinyl exhibited the same trend but with smaller deviations: the mean perfect square error increased from approximately 0.251 to 0.648 mm, and the mean squareness error increased from approximately 0.71% to 12.59% over the same feed rate range. In contrast, changes in pneumatic pressure did not produce a consistent monotonic variation in either positioning accuracy or contour repeatability.

Taken together, the results indicate that, once complete material separation is achieved, square-cutting fidelity is influenced more strongly by feed rate than by blade-loading pressure. Increasing feed rate primarily degraded cornering and squareness, whereas contour repeatability remained comparatively stable across repeated trials. Lower feed rates are therefore preferable when dimensional and angular fidelity are prioritized. The lower repeatability and squareness errors observed for vinyl further indicate that the material response can influence the geometric outcome, although the dominant trend with feed rate was consistent for both materials.

5.3. Material-Dependent Cutting Performance

Figure 9 presents the selected cutting conditions for five representative flexible sheet materials: PVC electrical tape, LDPE foam sheet, masking tape, polyimide film tape, and cloth tape. The results demonstrate that no single pressure–feed rate combination was suitable for all materials.

PVC electrical tape required relatively high pneumatic pressure because of its elastic recovery and adhesive friction. LDPE foam sheet and masking tape achieved stable cutting at moderate pressure, although excessive loading increased compression and tearing, respectively. Polyimide tape required the lowest pressure because its thin, relatively stiff film allowed clean penetration with minimal blade loading. Cloth tape required higher pressure and a lower feed rate because its fiber reinforcement increased cutting resistance.

Overall, the results demonstrate that cutting conditions depend strongly on material properties. Materials with higher cutting resistance, such as PVC electrical tape and cloth tape, required greater blade loading, whereas polyimide tape could be processed at lower pressure. These results highlight the benefit of programmable pneumatic blade loading for adapting cutting conditions to different flexible sheet materials.

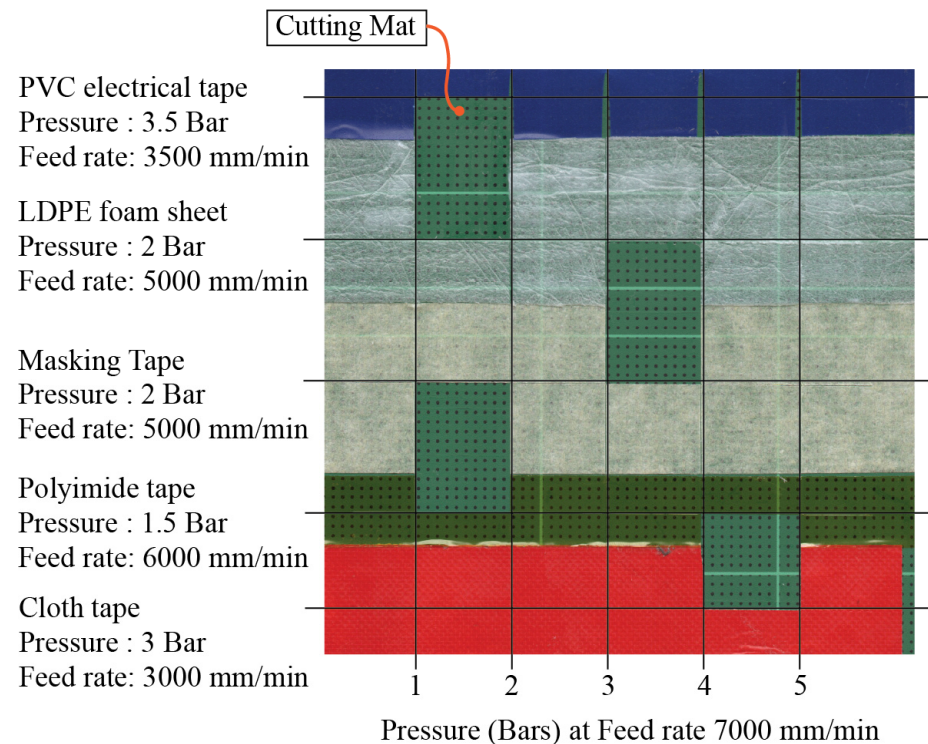


Figure 9. Material-dependent cutting response of five representative flexible sheet materials processed using the programmable pressure pneumatic knife CNC platform.

5.4. Manual–Machine Comparison

Figure 10 compares manual cutting with the programmable pressure pneumatic knife CNC platform using an S-shaped trajectory. The continuous curvature and changing direction provide a more demanding path than straight-line cutting.

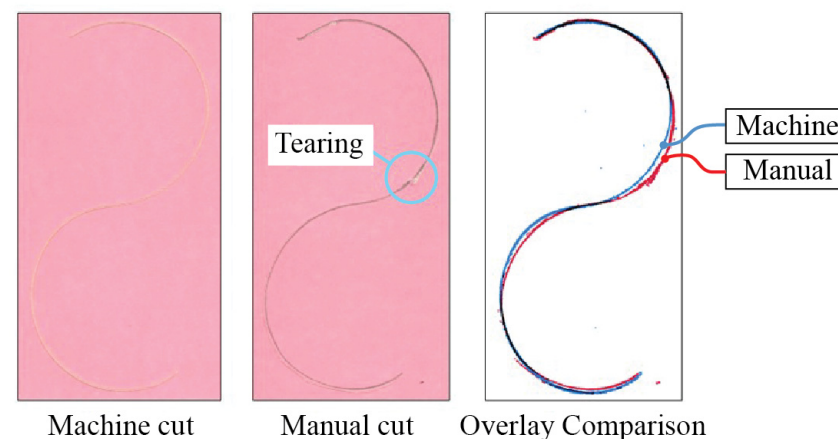


Figure 10. Comparison of machine and manual cutting for an S-shaped trajectory.

Manual cutting required 71 s and exhibited visible tearing, uneven edge quality, and local path deviations, particularly in curved regions. In comparison, the CNC platform completed the same trajectory in 3.65 s, corresponding to a 19.5-fold increase in cutting speed and a 94.9% reduction in processing time. The machine cut sample also showed a smoother and more continuous cutting path with fewer visible defects.

The trajectory overlay showed greater local deviation in the manually cut sample, particularly around curved transitions, while the machine cut path more closely followed the programmed trajectory. These results demonstrate improved processing speed, path consistency, and reduced operator dependence using the CNC platform.

5.5. Demonstration Applications

5.5.1. Kirigami Gripper

The kirigami gripper demonstrated the capability of the programmable pressure pneumatic knife CNC platform to fabricate dense, repeated slit patterns for soft robotic structures, as shown in Figure 11. The pattern was cut at 3 bar and 3000 mm min^{-1} in 13.44 s, producing clean and continuous cuts with complete separation and acceptable edge quality. The selected pressure–feed rate condition provided sufficient blade penetration without excessive crushing.

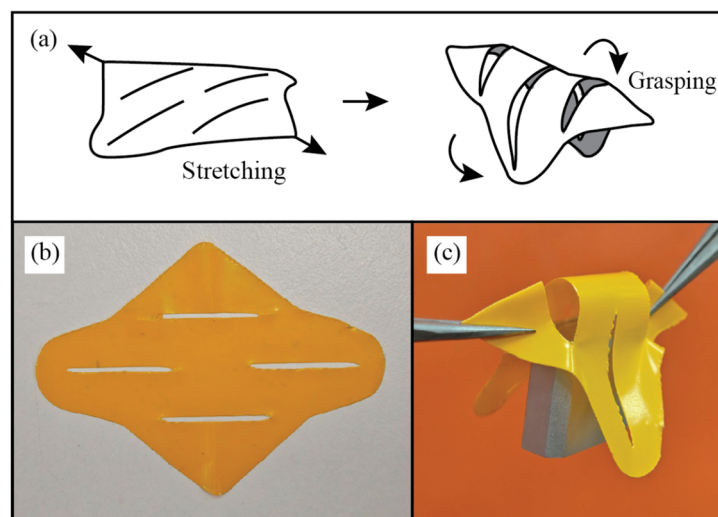


Figure 11. Fabrication and validation of the kirigami gripper: (a) Schematic of the deformation mechanism under stretching, (b) Fabricated sheet with slit patterns, and (c) Demonstration of object grasping.

A geometry-dependent limitation was observed when slit features were too small or adjacent cut lines were placed too closely, resulting in tearing during cutting or handling. This tearing was associated with reduced ligament width between neighboring cuts. The results indicate that reliable kirigami fabrication depends on both appropriate cutting conditions and sufficient feature spacing.

5.5.2. Flexible Bag

In Figure 12, the flexible bag demonstrated the capability of the programmable pressure pneumatic knife CNC platform to cut closed contours and internal features in thin flexible sheets for soft pneumatic structures.

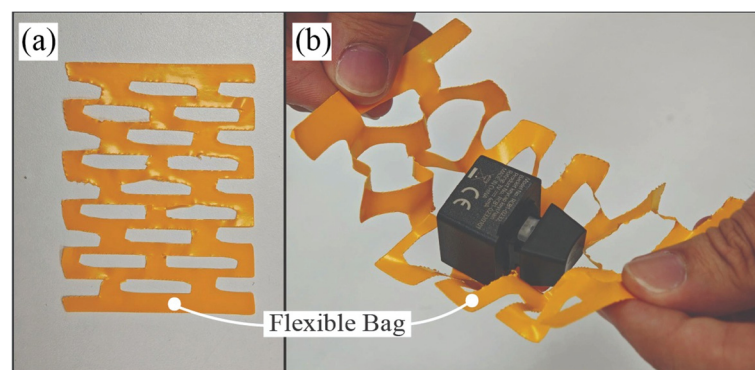


Figure 12. Fabrication and demonstration of the flexible bag: (a) Fabricated pattern in flat state, and (b) Demonstration holding an object.

The pattern was cut at 3 bar and 3000 mm min^{-1} in 44.02 s. The main outer contour was successfully cut under this condition. However, incomplete cutting occurred around small curved holes because the passive blade orientation became less suitable for tight-radius paths. The remaining uncut regions required manual removal. These results demonstrate effective cutting of large flexible-bag contours, while small-radius internal features remain challenging for the passive knife. Toolpath curvature and hole size should therefore be considered when designing such patterns.

5.5.3. Silicone Breast Scar Sheet

In Figure 13, the silicone breast scar sheet demonstrated the capability of the programmable pressure pneumatic knife CNC platform to cut soft, highly flexible elastomeric sheets for wearable applications.

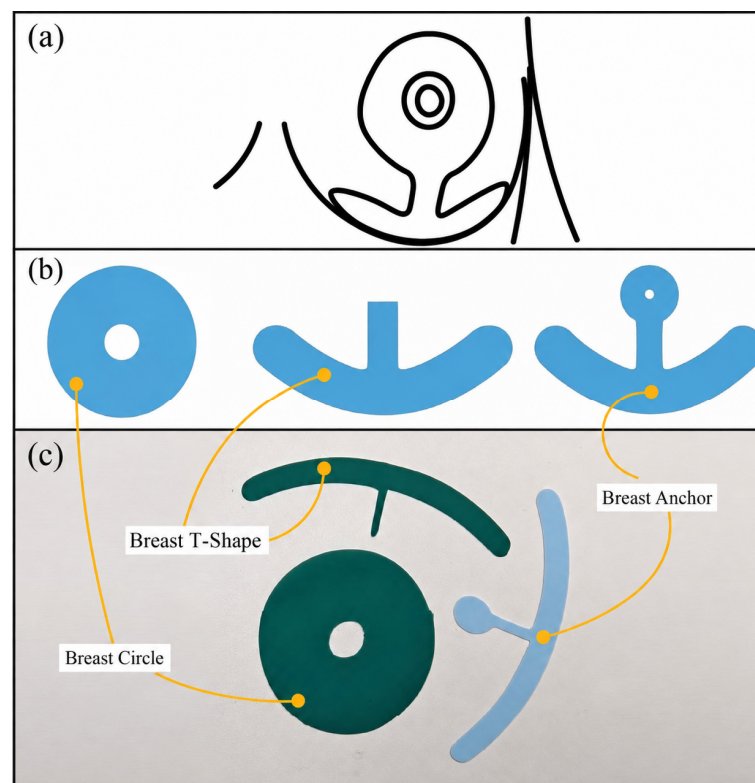


Figure 13. Fabrication of a silicone breast scar sheet: (a) Application, (b) Component design, and (c) Fabricated silicone scar-sheet components.

The sheet was cut at 3 bar and 2000 mm min^{-1} in 15.10 s, producing effective material separation. However, deformation of the highly flexible silicone caused path deviation during curved cutting, particularly around small-radius and fine-detail features. The results demonstrate the suitability of the platform for silicone sheets, while highlighting the importance of material fixation, feature size, and toolpath curvature for maintaining geometric accuracy.

6. Discussion

6.1. Role of Programmable Pneumatic Pressure

The results demonstrate that programmable pneumatic pressure is a central design feature of the cutting platform. Pressure defines the nominal blade-loading condition and directly affects penetration and cutting stability. Insufficient pressure can result in skimming or no-touch behavior, whereas excessive pressure increases frictional drag,

material deformation, and tearing. Stable cutting therefore requires an appropriate pressure selected based on the material and toolpath geometry.

In the present system, 1 bar corresponds to approximately 7.85 N of nominal normal blade-loading force, providing a practical interpretation of pressure as a controlled loading variable. However, the system does not implement closed-loop force feedback; it provides repeatable pressure-programmed nominal loading conditions. The experimentally determined pressure–feed rate relationships provide reproducible process conditions rather than direct force-controlled cutting.

The results further support the hybrid position–force concept of the platform. The X–Y axes control the toolpath and feed rate, while the pneumatic subsystem controls nominal blade loading. This separation enables cutting conditions to be adjusted independently of toolpath motion, which is particularly beneficial for flexible sheet materials with different cutting responses.

6.2. Material-Dependent Cutting Behavior

A major finding of this study is that flexible sheet cutting is strongly material dependent. Differences in stiffness, thickness, adhesion, compressibility, elastic recovery, and reinforcement resulted in distinct pressure–feed rate conditions for stable cutting. Thus, flexible sheet cutting is better treated as a process window problem rather than a single-parameter cutting task.

Paper-based materials were more susceptible to tearing under excessive pressure, consistent with their lower tear resistance and tendency toward local deformation [8]. In contrast, vinyl- and PVC-based materials required higher pressure to maintain blade penetration because of their compliant structure and elastic recovery. The material-specific trials further showed that polyimide tape was successfully processed at lower pressure. LDPE foam sheet required moderate blade loading to limit compression, whereas cloth-backed tape required higher pressure and lower feed rate because of its fiber reinforcement.

Toolpath geometry also affected the operating window. Straight-line and square cutting did not always produce the same optimal conditions because direction changes and corners alter blade orientation and local material deformation. The passive self-aligning knife can accommodate gradual direction changes, but sharp corners and small-radius features may introduce temporary blade misalignment.

These findings indicate that a universal cutting condition is unlikely to be reliable across different materials and geometries. Instead, recommended pressure–feed rate ranges should be defined according to material and toolpath characteristics. The platform provides a practical basis for developing such process window data through programmable pressure and feed rate control.

6.3. Advantages over Manual Cutting

The manual–machine comparison demonstrates the practical advantage of the programmable pressure CNC approach. Manual cutting required 71 s for the S-shaped trajectory and exhibited visible tearing, uneven edge quality, and local path deviations. In comparison, the CNC platform completed the same trajectory in 3.65 s, corresponding to a 19.5-fold increase in cutting speed and a 94.9% reduction in processing time.

The machine cut path also showed greater trajectory consistency and fewer visible defects, reflecting the benefits of controlled feed motion, repeatable toolpath execution, and programmable blade loading. This improvement is particularly relevant for flexible sheet components in which local geometric accuracy can affect functional performance. However, curved cutting paths still produced some deviation because the passive knife must follow

changes in cutting direction. Further improvement could therefore involve curvature-aware toolpath planning and active blade orientation control for tight-radius features.

6.4. Implications for Automated and Robotic Fabrication

The study demonstrates how a simple mechanical cutting process can be transformed into a programmable manufacturing system through the separation of toolpath control from blade-loading control. Pressure, feed rate, tool position, and knife engagement can be specified within a unified CNC workflow rather than adjusted manually.

This architecture is particularly relevant to soft robotic and wearable-device fabrication [3,27]. Flexible sheet components often require precise and repeatable cutting, and material-specific cutting conditions can reduce trial-and-error during fabrication. The Klipper-based control architecture also provides an open and transparent platform in which machine parameters, toolpaths, and pneumatic actuation can be modified and documented [14].

The results further indicate that reliable flexible sheet cutting depends on the interaction of machine parameters, material behavior, and toolpath geometry rather than on a single control variable. The platform therefore provides a practical basis for systematic process characterization and future extensions such as camera-based inspection and adaptive process control.

6.5. Limitations and Future Work

The platform has several limitations. First, blade loading is pressure-programmed rather than force-controlled, so the actual cutting force may vary with actuator dynamics, blade orientation, material compliance, and surface conditions. Future work should incorporate force sensing at the toolhead.

Second, the passive blade alignment becomes less effective for tight-radius curves and dense features, as observed in the flexible bag demonstration. Active blade orientation control and curvature-dependent feed rate adjustment could improve cutting accuracy [30]. Material fixation also remains important for flexible and highly compliant sheets.

Finally, the material-specific experiments were not full operating window studies for every substrate, and the image-based evaluation provided limited quantitative metrology. Future studies should expand pressure–feed rate mapping and include dimensional accuracy, edge quality, force measurement, and repeatability.

7. Conclusions

This work developed and experimentally validated a pneumatic knife CNC platform with programmable pressure for flexible sheet cutting. Results demonstrated that cutting quality is governed by a material- and geometry-dependent pressure–feed rate process window, with optimal conditions varying among different flexible sheet materials. The platform achieved up to a 99% clean cut rate for paper and 100% for vinyl while reducing the processing time of an S-shaped trajectory by 94.9% compared with manual cutting. Repeated square-cutting tests confirmed the geometric repeatability of the platform. Positioning errors were 0.223–0.805 mm for paper and 0.183–0.597 mm for vinyl, with contour repeatability of 0.150–0.346 mm and 0.076–0.193 mm, respectively. Lower feed rates generally improved squareness and corner fidelity. Fabrication of a kirigami gripper, flexible pneumatic bag, and silicone scar sheet further highlighted the platform’s applicability to soft robotic, pneumatic, wearable, and healthcare device fabrication. Overall, this work establishes programmable pneumatic blade loading as an effective and reproducible approach for flexible sheet cutting, enabling controllable digital fabrication and systematic parameter selection.

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